

The Organic Chemistry of Friction Control

Lubrication is not physical separation.
It is a molecular reaction.



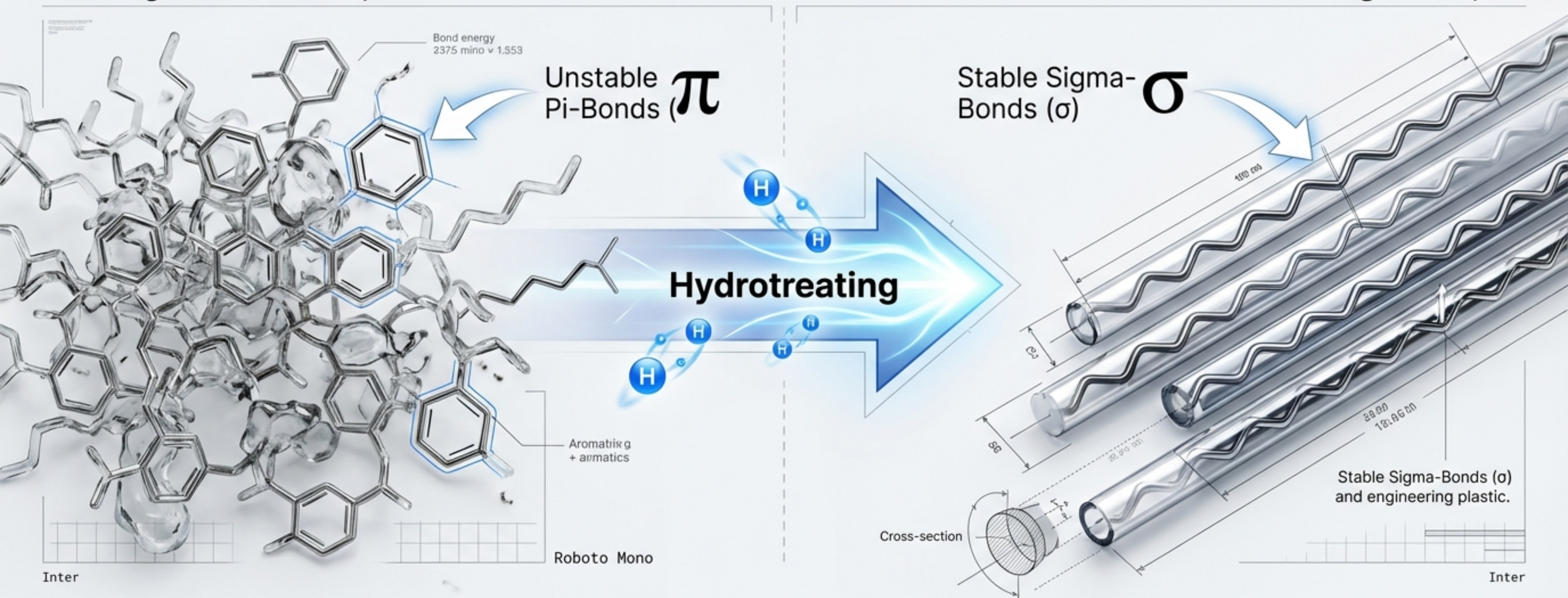
COMPOSITION ANALYSIS

- Base Oil: 70–99% (The Backbone)
- Additives: 1–30% (The Intelligence)
- Molecular Range: C15 – C50

From Refining Nature to Architecting Molecules

Refining Nature / Groups I-III

Architecting / Group IV

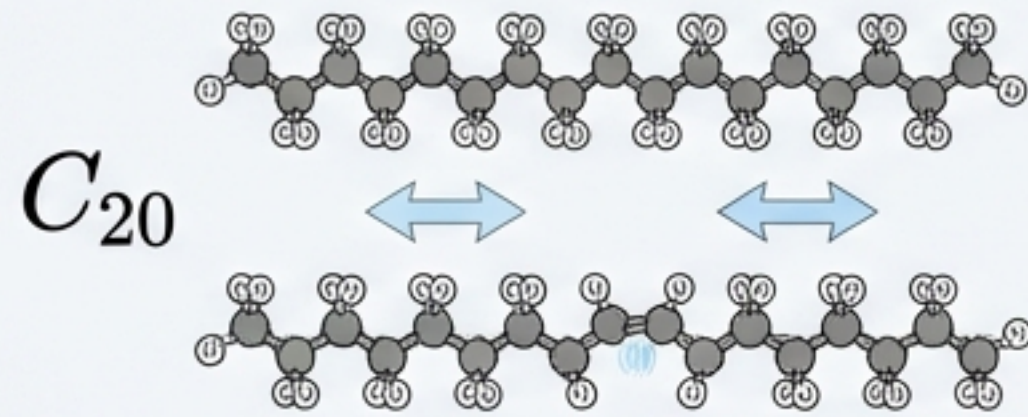


MINERAL (Natural Chaos):
Saturated Hydrocarbons + Aromatics.
Prone to sludge.

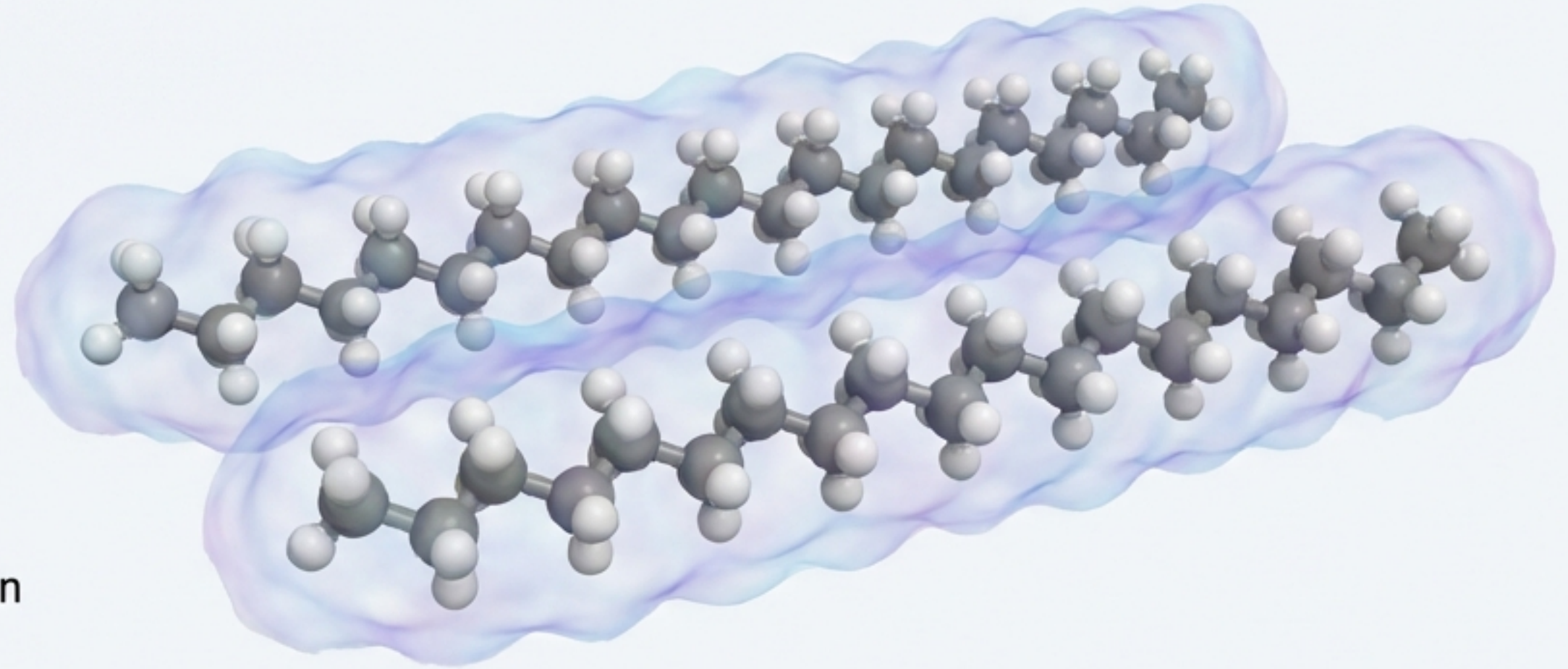
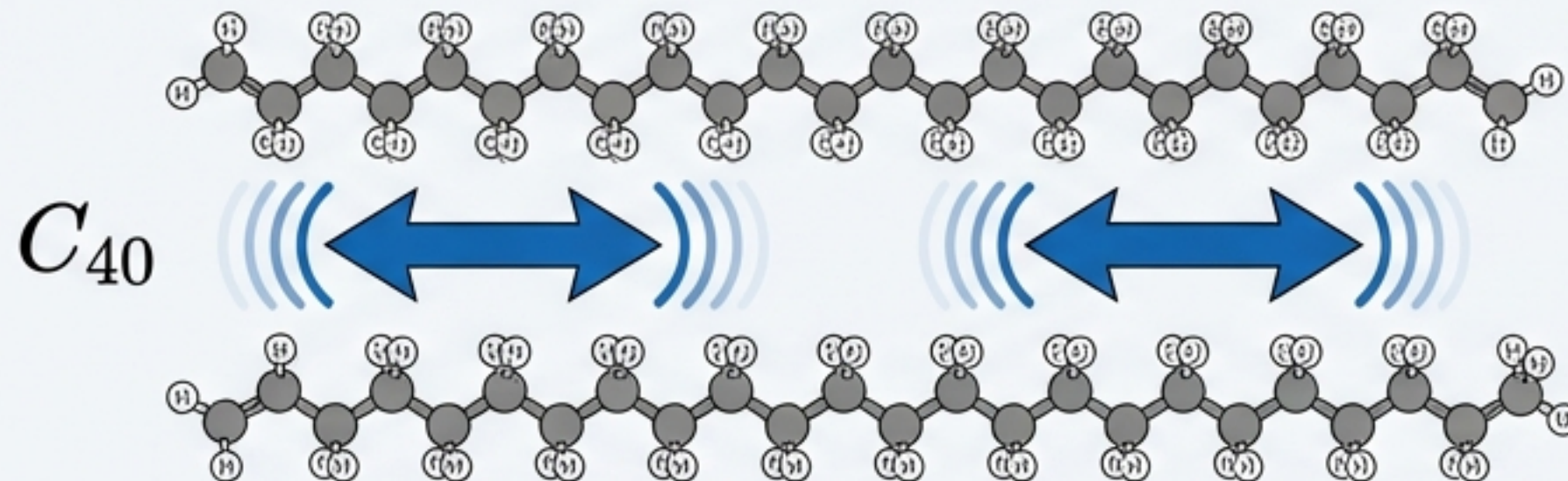
SYNTHETIC (Engineered Order):
Pi-bonds broken into stable
Sigma-bonds. Oxidation resistant.

The Physics of Stickiness: Van der Waals Forces

The Mechanism



Increasing Attraction



THE MECHANISM

- No hooks. No glue.
- Fleeting electrical dipoles created by electron wobble.
- Longer Chains = More Surface Area = Stronger Attraction (Viscosity).

Data Point: Attraction force scales roughly with chain length squared.

The Viscosity Paradox: Spaghetti vs. Rice

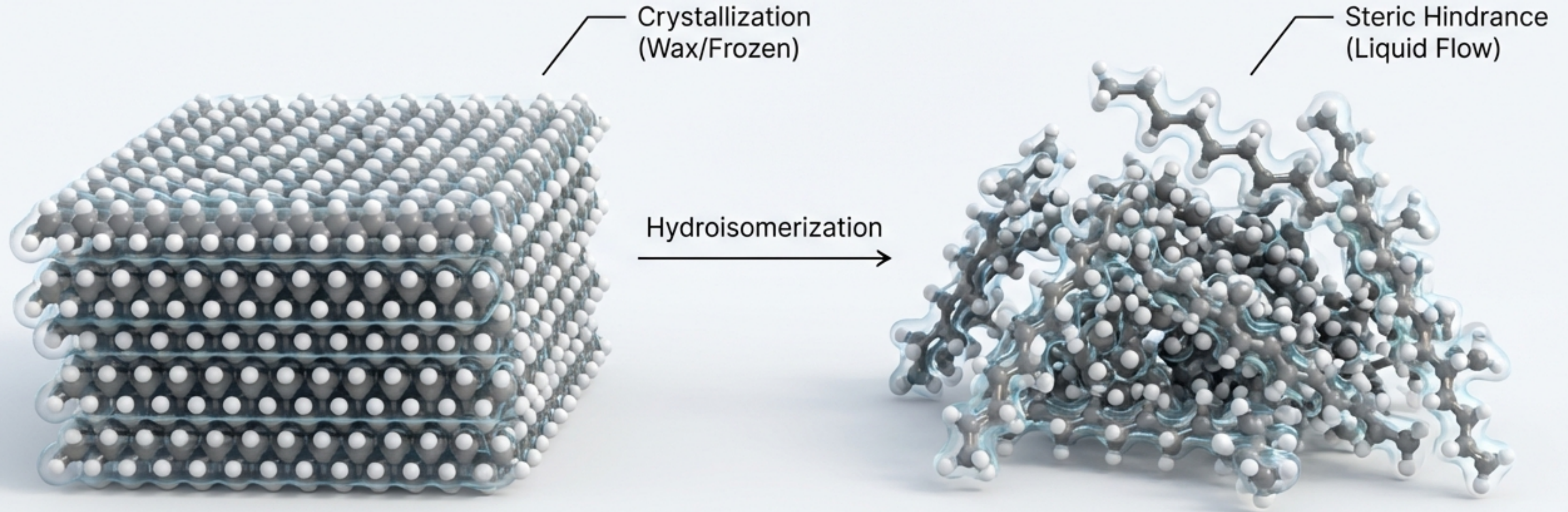


Viscosity Index (VI) is the measure of a molecule's ability to maintain the 'tangle' as heat adds energy.

Engineering Cold Flow: The Steric Hindrance Solution

Straight Chains / n-Paraffins

Branched Chains / iso-Paraffins



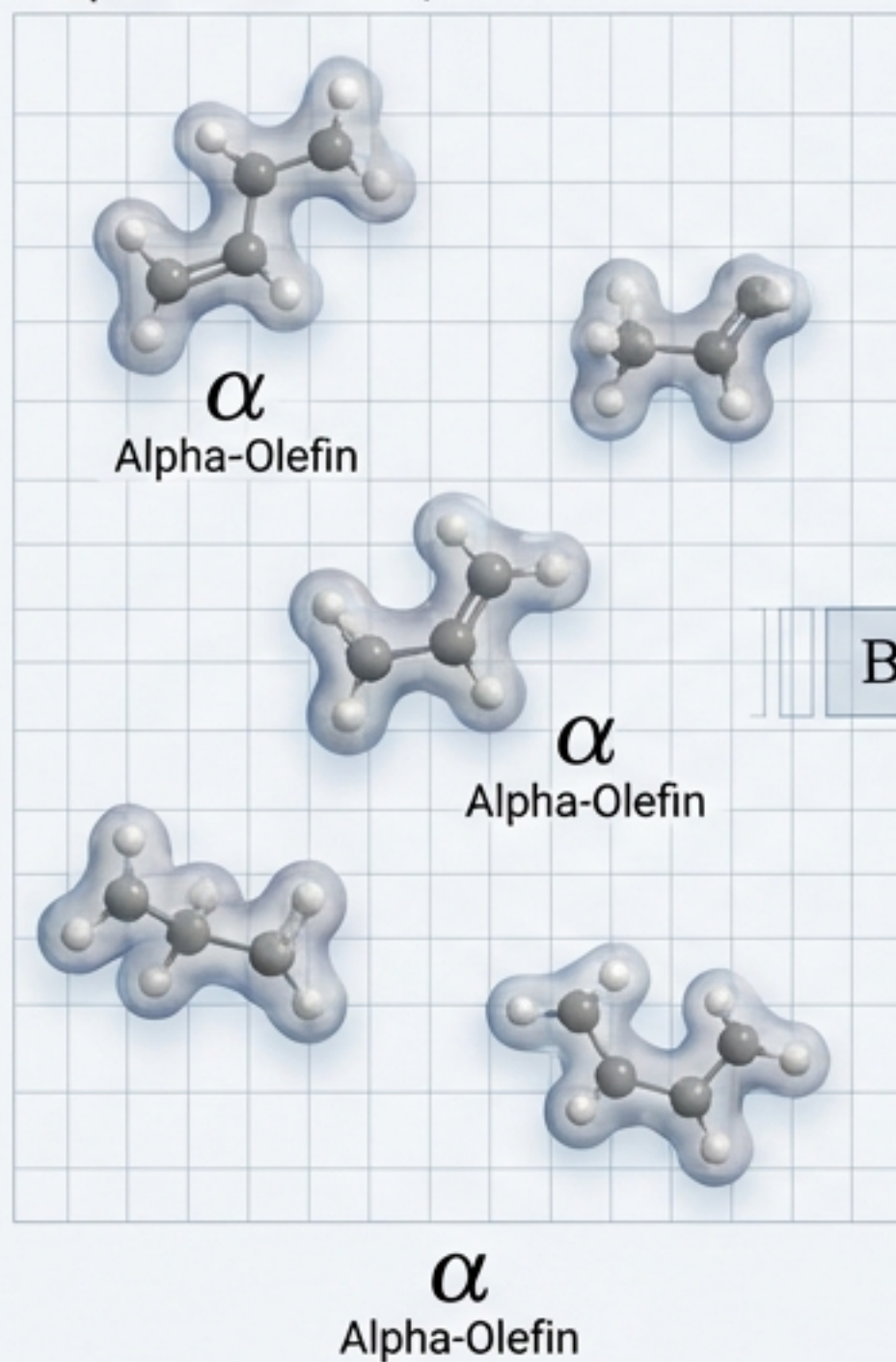
Hydroisomerization rearranges the carbon backbone. The branches act as physical bumps, preventing the molecules from stacking into a solid wax at low temperatures.

Group IV PAO: The Designer Molecule

THE PAO ADVANTAGE

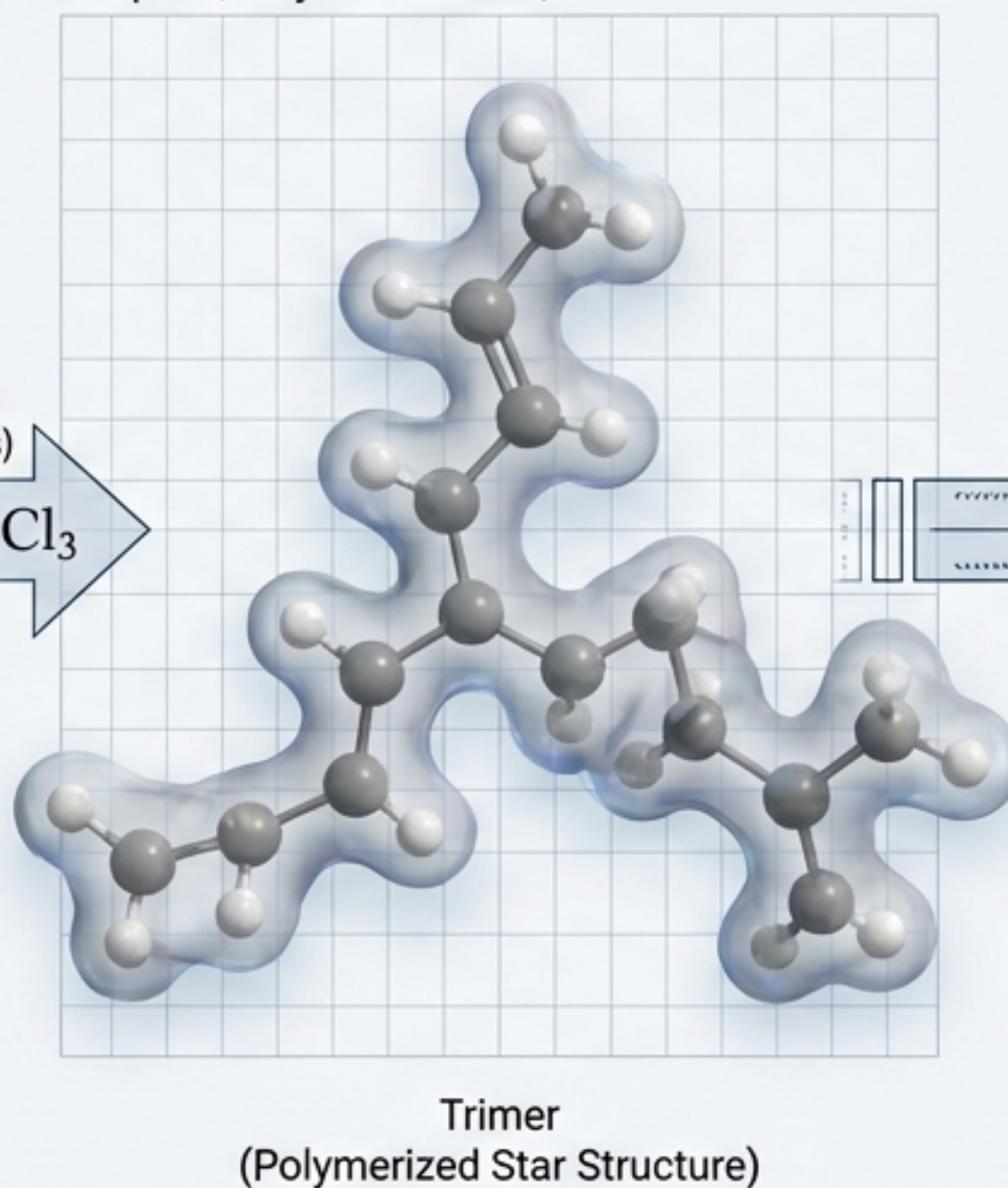
- Zero Sulfur, Zero Aromatics.
- Tuned Structure: Every molecule is identical.
- Elimination of light ends (Volatility) and heavy ends (Wax).

Step 1 (Monomers)

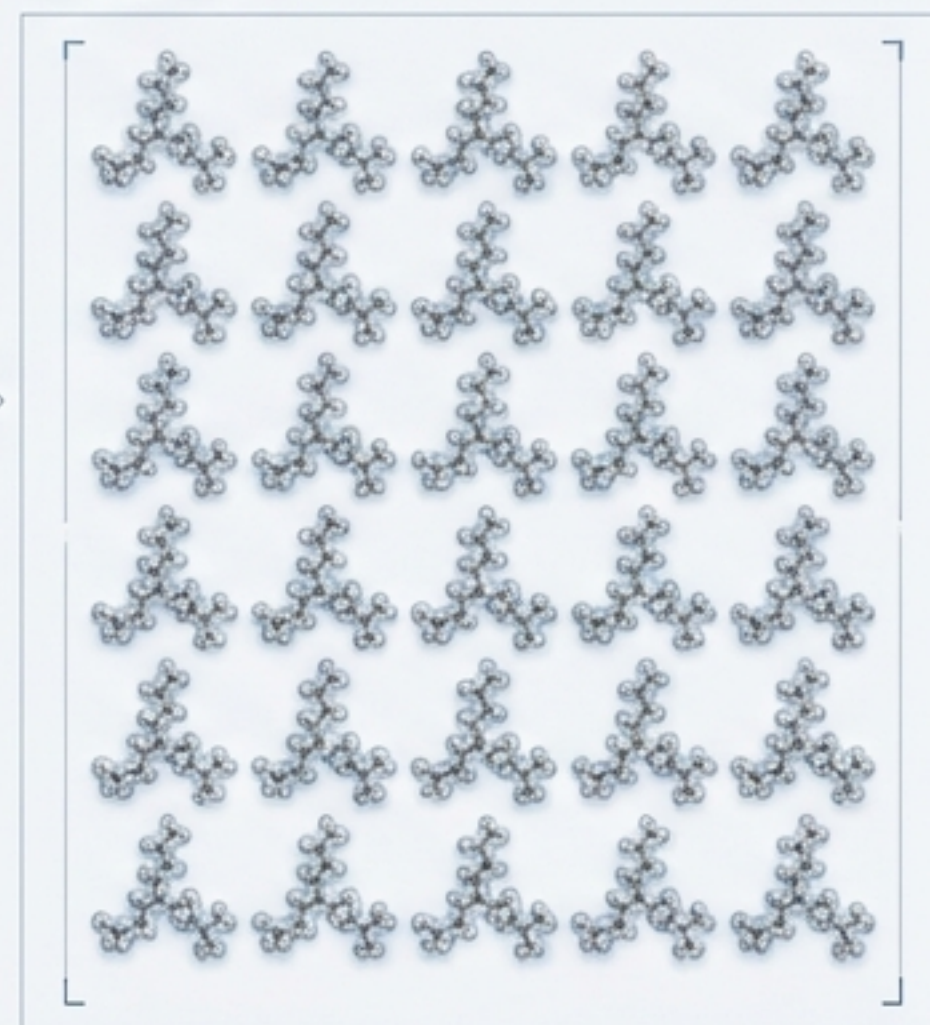


Step 2
(Catalysis)
 BF_3 or AlCl_3

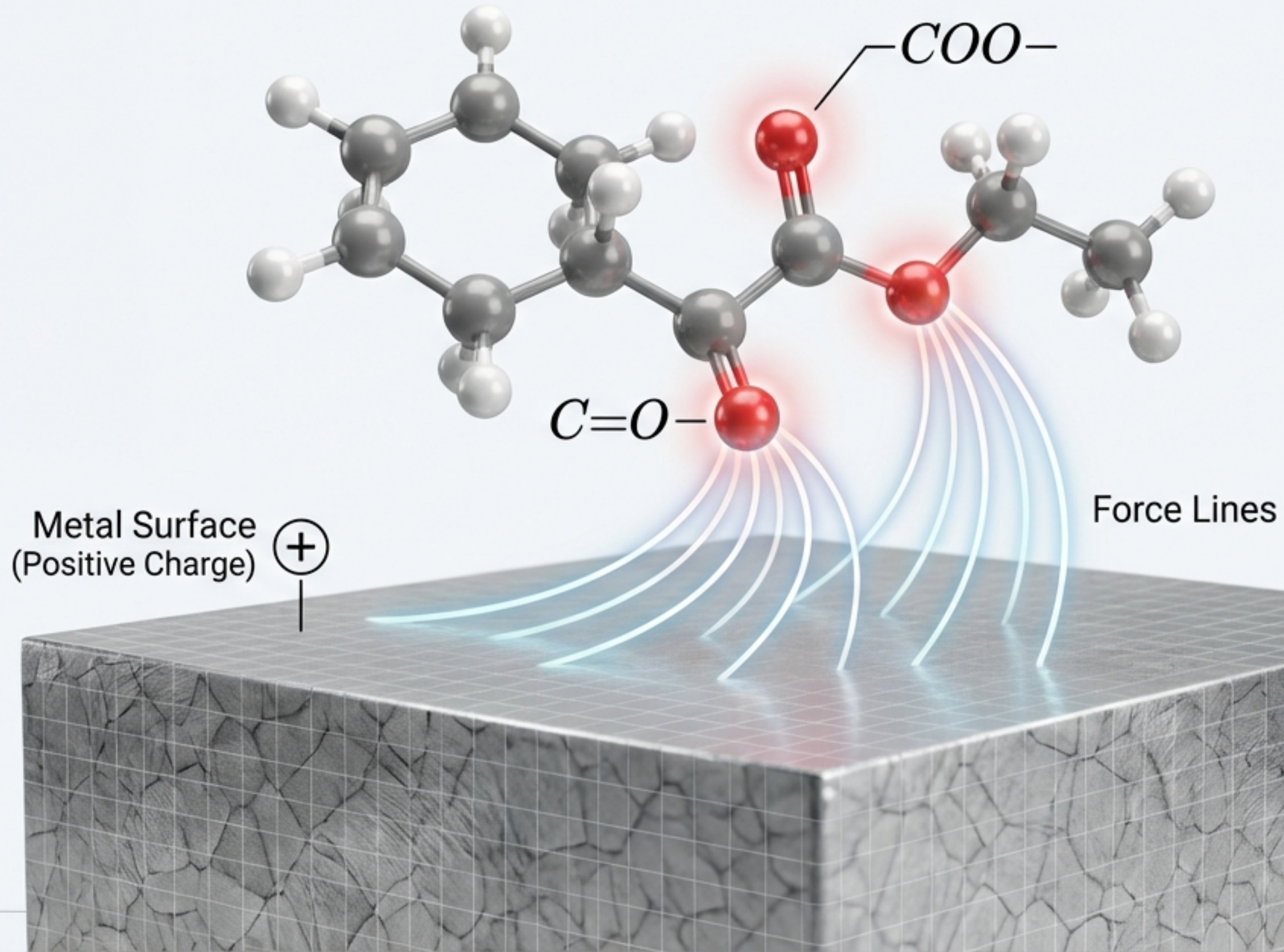
Step 3 (Polymerization)



The Result

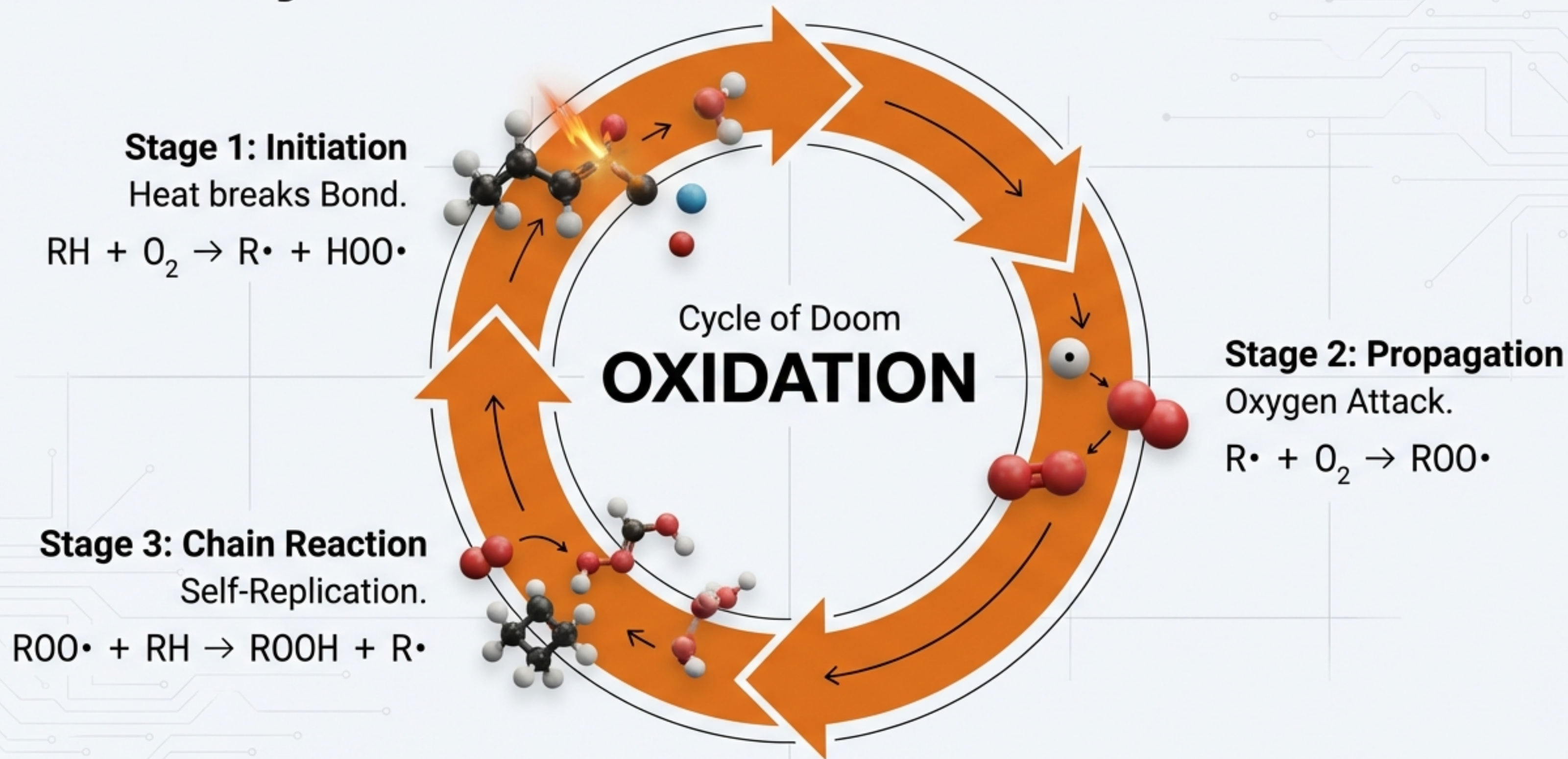


Group V Esters: Their Magnetic Personality



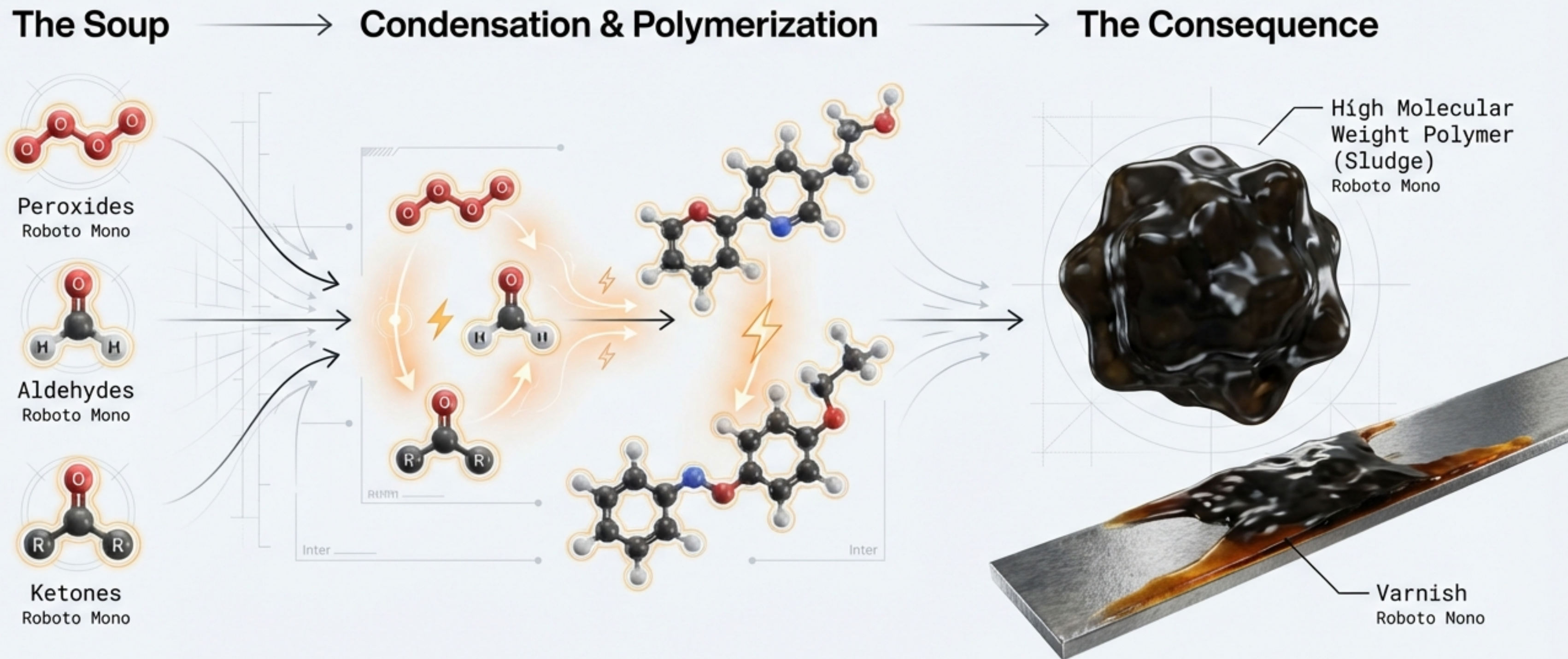
- Mechanism: Polarity / Dipole Moment
- Bond Energy: 360 kJ/mol (High Thermal Stability)
- Result: Wettability & Natural Solvency

The Enemy: The Radical Chain Reaction



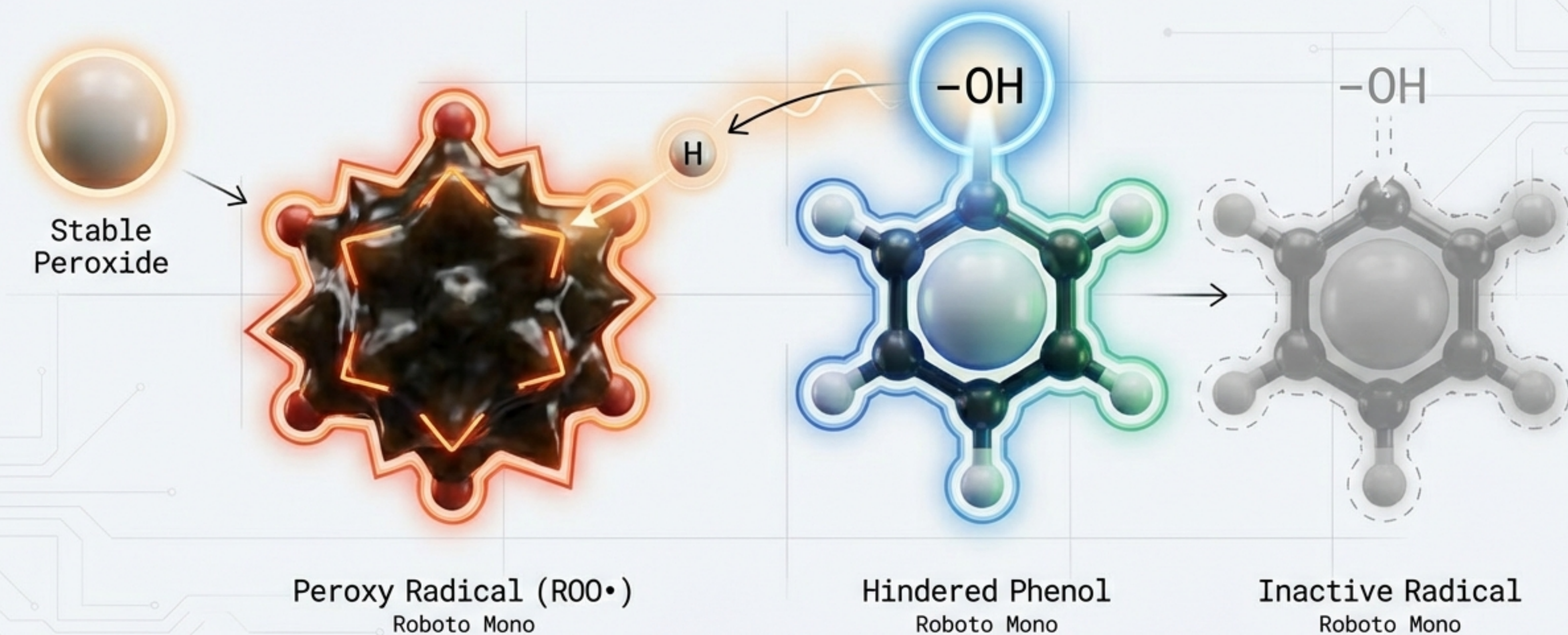
Unchecked, this cycle destroys the base oil exponentially.

From Fluid to Sludge: The Polymerization of Death



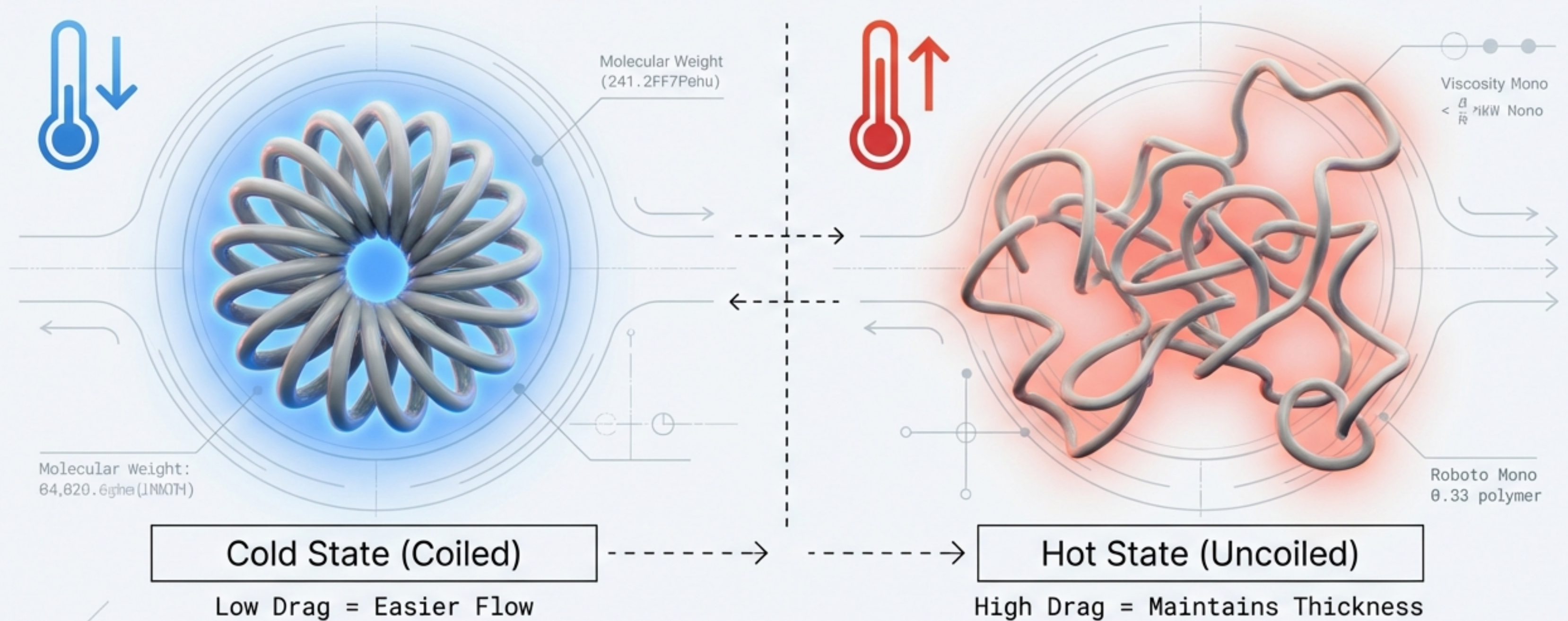
“Oxidation byproducts link up to form insoluble solids, acting as the “blood clots” of the engine.”

The Shield: Antioxidants as Radical Hunters



The Sacrificial Lamb: The antioxidant donates a Hydrogen to neutralize the radical, dying so the oil can live.

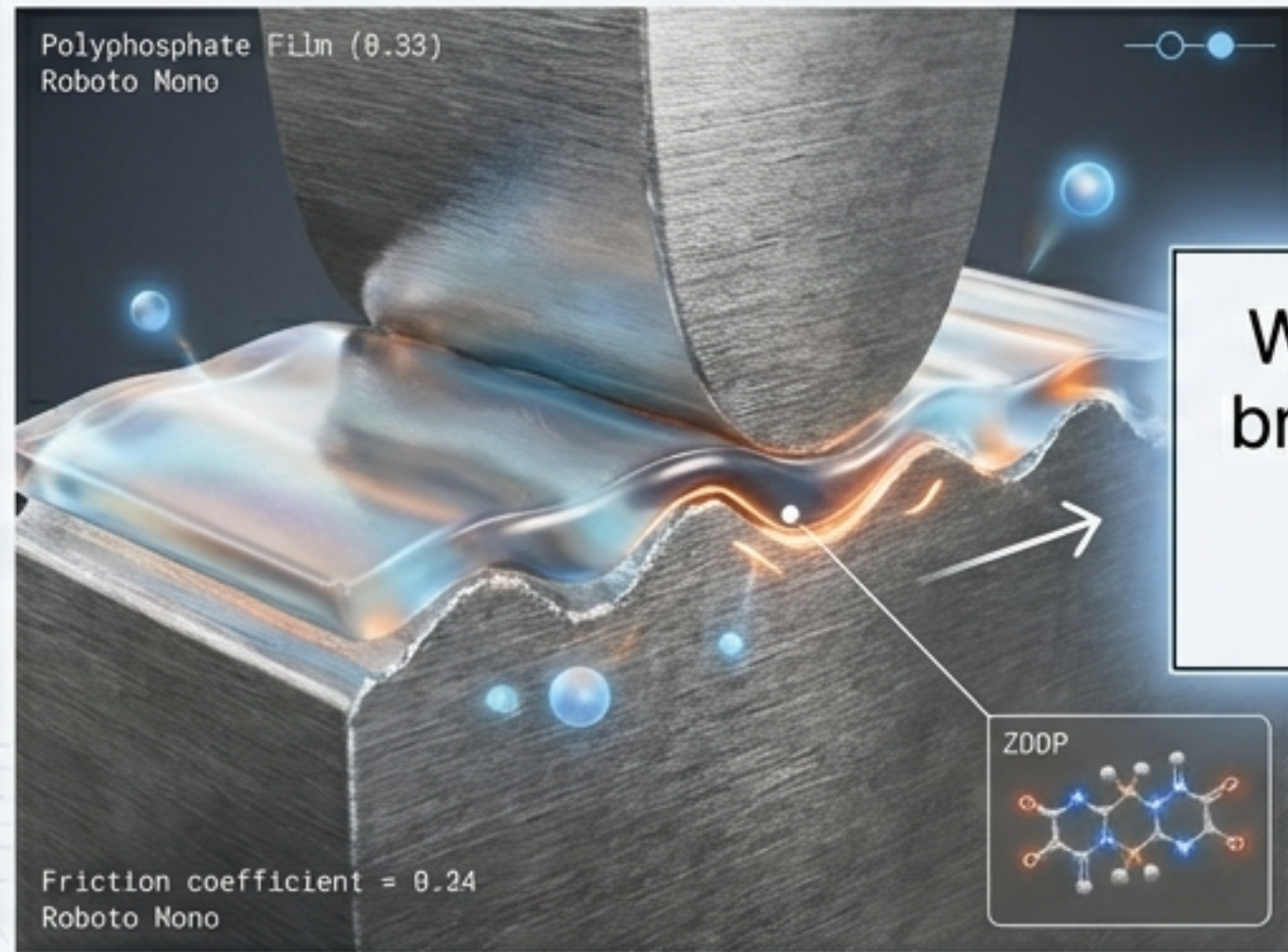
The Shape-Shifter: Viscosity Index Improvers



Acts like a thermal spring: expanding to create internal friction when the oil naturally wants to thin out.

The Last Line of Defense: AW vs. EP

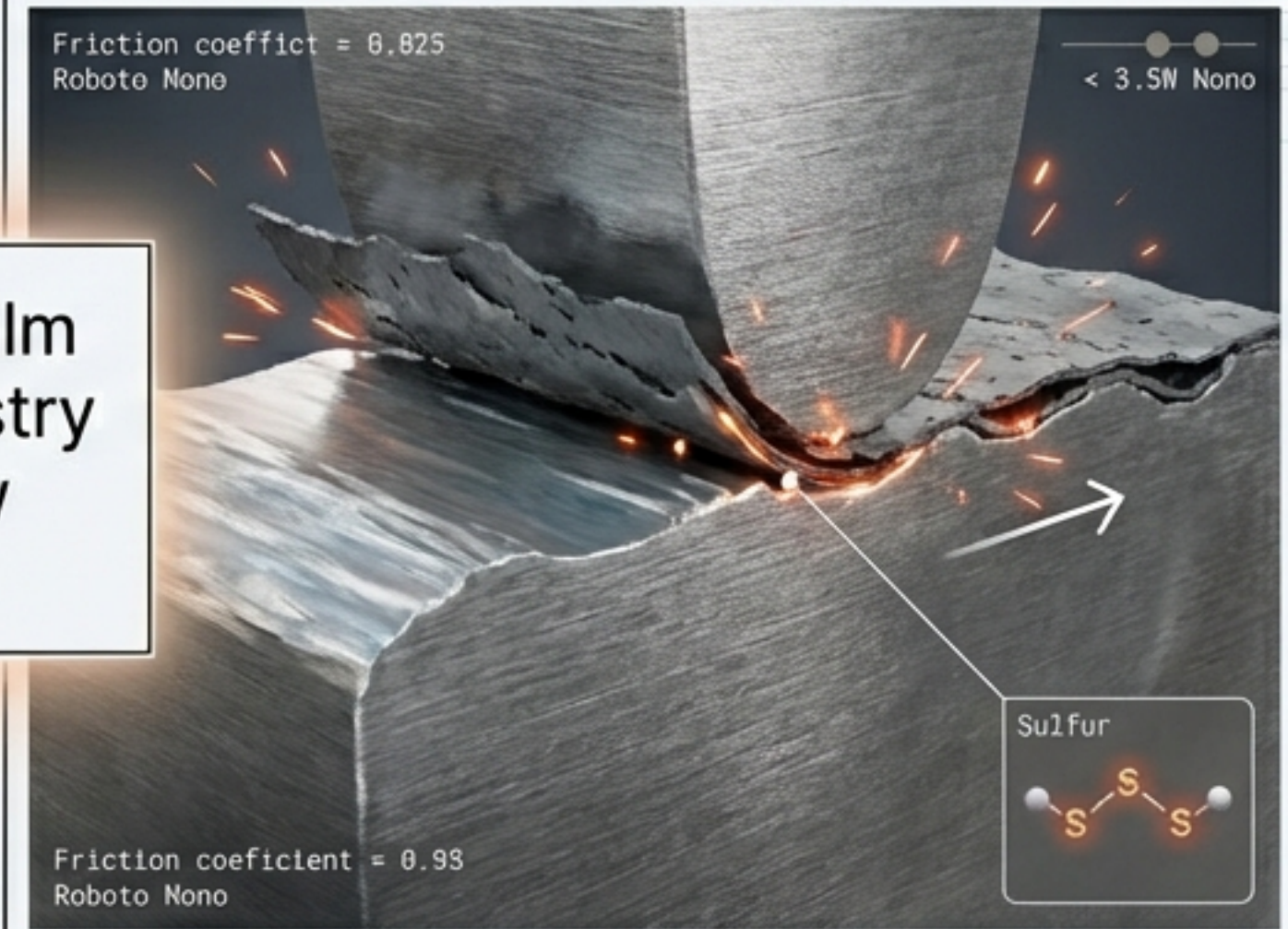
Anti-Wear (ZDDP)



Polyphosphate Film (Glass-like)

Plastic deformation.
Protective coating.

Extreme Pressure (Sulfur)



Metal Sulfide Film (Sacrificial)

Controlled shearing.
Prevents welding.

When the oil film
breaks, chemistry
builds a new
surface.

The Cleaners: Scrubbing vs. Enveloping

Detergents (The Neutralizer)

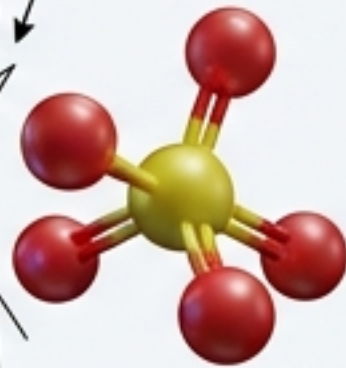
CHEMICAL NEUTRALIZATION

pH Balance:
7.0 (Neutral)



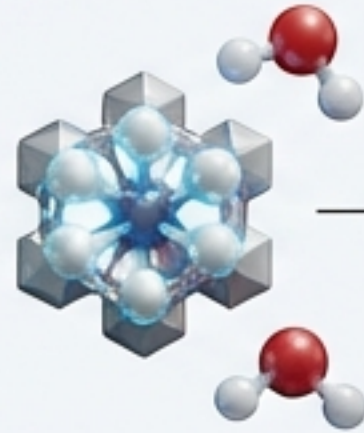
Metallic Salt
(Calcium)

pH Balance:
(H_2SO_4)



Sulfuric Acid
(H_2SO_4)

=

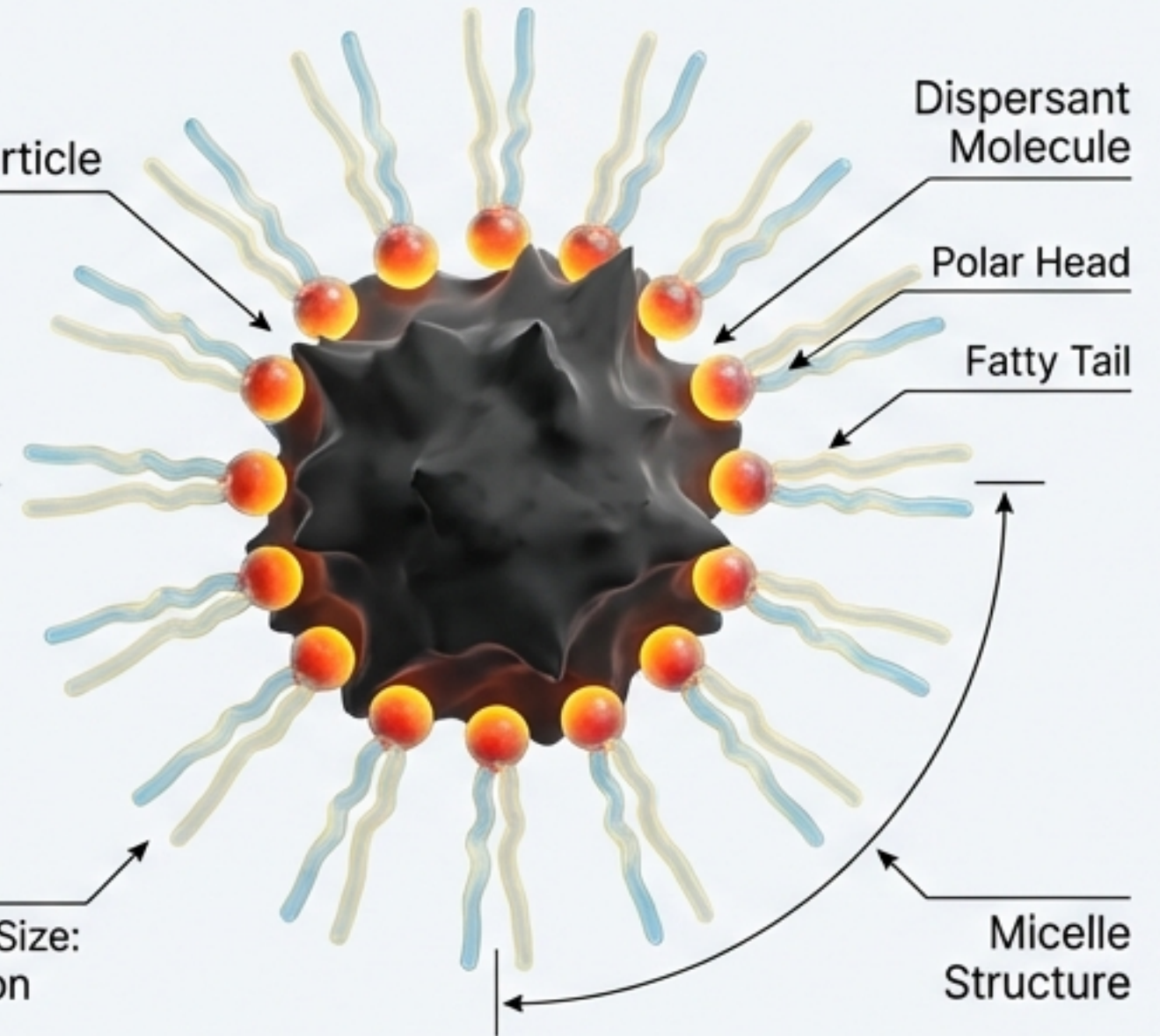


Neutralized
Salt + Water

Dispersants (The Envelope)

PHYSICAL ENCAPSULATION

Soot Particle

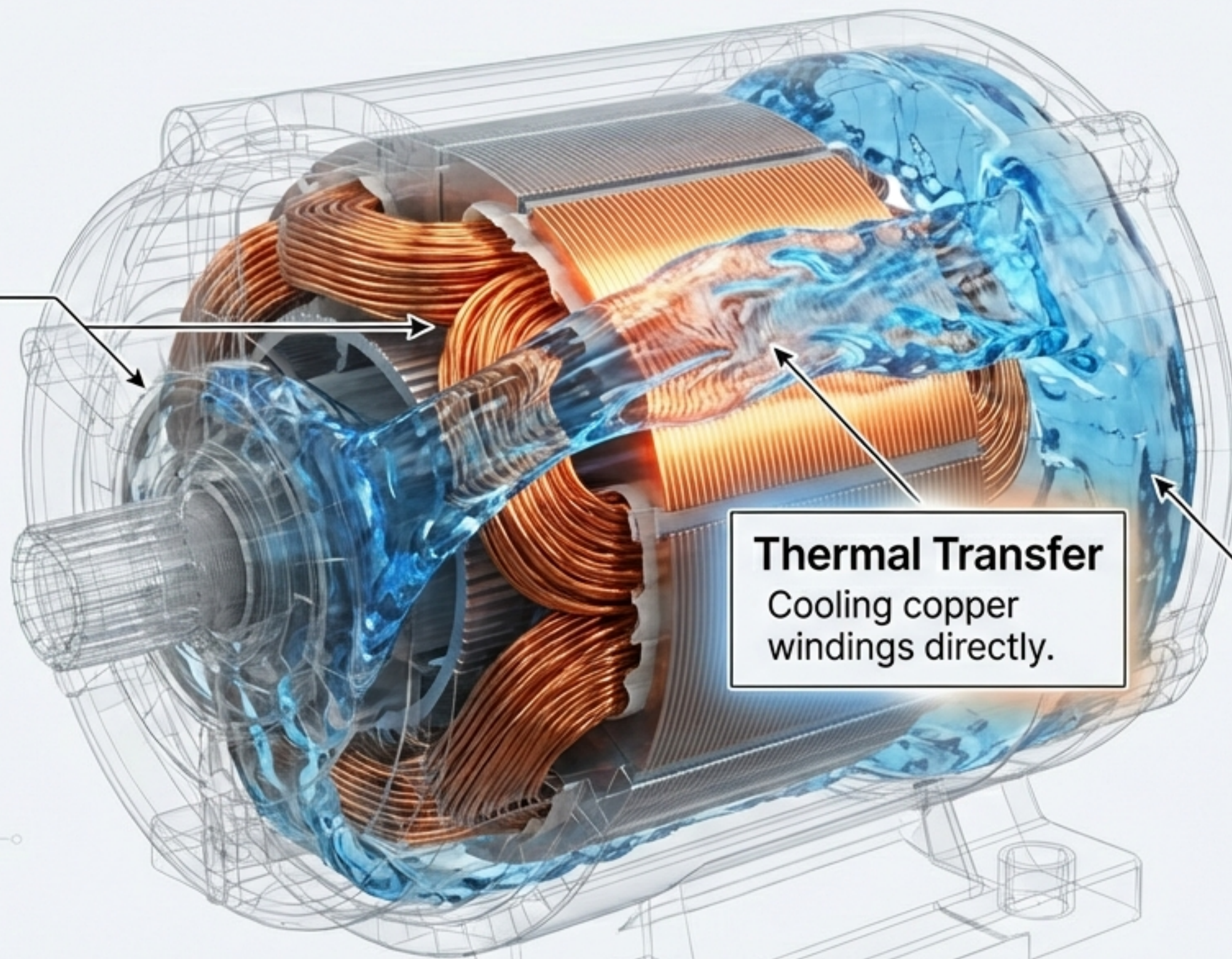


Detergents fight acid corrosion. Dispersants keep soot floating to prevent sludge.

The Future of Fluids: Electrification & Sustainability

Dielectric Properties

Must insulate high voltage. No conductive metal additives.



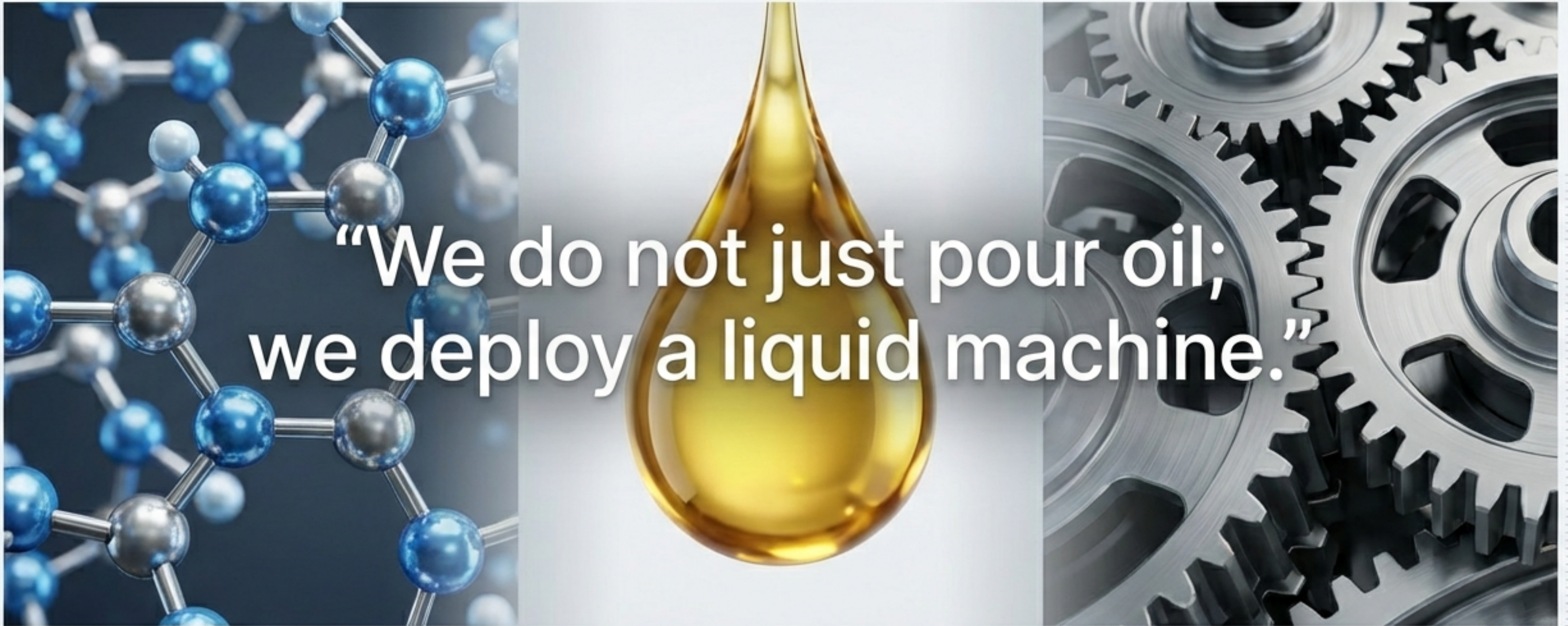
Thermal Transfer

Cooling copper windings directly.

Bio-Esters

Modified natural triglycerides for carbon neutrality.

The Ultimate Fusion of Physics & Chemistry



The Chemistry

The Fluid

The Physics

Lubrication is Molecular Engineering.